Restaurant Data Analysis Project

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Project Type: Exploratory Data Analysis (EDA)

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn

Environment: Jupyter Notebook

Introduction

The objective of this project is to perform an exploratory data analysis (EDA) on a restaurant dataset in order to understand patterns in cuisines, city distribution, ratings, pricing, and online delivery services. The goal of the analysis is to extract meaningful insights from the dataset and identify trends that influence restaurant popularity and customer preferences.

The analysis was conducted using Python in a Jupyter Notebook environment. Several data science libraries such as **Pandas, NumPy, Matplotlib, and Seaborn** were used for data manipulation, visualization, and statistical interpretation.

Through this project, I aimed to develop a deeper understanding of real-world restaurant data and apply fundamental data analysis techniques including data cleaning, aggregation, visualization, and interpretation.

Tools and Libraries Used

The following Python libraries were used during the project:

Pandas

Pandas was used for:

- Data loading
- Data cleaning
- Data transformation
- Aggregation and grouping

NumPy

NumPy was used for:

- Numerical computations
- Efficient array operations

Matplotlib

Matplotlib was used to create:

- Bar charts
- Distribution plots
- Visual summaries of restaurant statistics

Seaborn

Seaborn was used for advanced visualizations and better graphical representation of trends in the dataset.

Dataset Overview

The dataset contains information about various restaurants including:

- Restaurant names
- Cities
- Cuisines served
- Ratings
- Price range
- Online delivery availability
- Table booking availability
- Votes received

Each record in the dataset represents a single restaurant with its associated attributes.

Some restaurants serve **multiple cuisines**, which required additional preprocessing to properly analyse cuisine distribution.

Data Preparation and Cleaning

Before performing analysis, several preprocessing steps were performed.

Handling Multiple Cuisines

Many restaurants listed more than one cuisine separated by commas. To properly analyze cuisine popularity:

- The cuisine column was split into individual cuisines.
- Each cuisine was counted separately.

This allowed a more accurate measurement of the popularity of each cuisine type.

Data Validation

Basic checks were performed to ensure:

- No missing critical values
- Proper formatting of columns
- Numerical columns were correctly interpreted

This ensured that the analysis results were reliable.

Level 1 Analysis

Task 1: Top Cuisines

The first task was to determine the **three most common cuisines** served by restaurants in the dataset.

Methodology

1. The cuisine column was split into individual cuisine entries.
2. Each cuisine was counted across the dataset.
3. Frequencies were calculated.
4. Percentages were computed to determine their overall representation.

Findings

The three most common cuisines identified were:

- **North Indian**
- **Chinese**
- **Fast Food**

These cuisines appeared most frequently across the dataset.

Interpretation

This indicates that customer demand is highly concentrated around these cuisine categories. Restaurants tend to offer these cuisines either individually or in combination with others. Their popularity likely reflects customer preferences and market demand in urban areas.

Other cuisines were present but appeared significantly less frequently.

Task 2: City Analysis

This section focused on understanding restaurant distribution across different cities.

Objectives

1. Identify the city with the **highest number of restaurants**
2. Calculate the **average rating of restaurants for each city**

3. Determine which city has the **highest average rating**

Methodology

The dataset was grouped by city using Pandas' groupby function. Aggregation techniques were then used to compute:

- Number of restaurants per city
- Mean rating per city

Findings

The analysis showed that some cities have a significantly larger concentration of restaurants compared to others. These cities are likely major metropolitan areas with larger populations and higher demand for dining services.

The average rating analysis revealed that certain cities consistently maintain higher customer satisfaction levels, which could indicate:

- Better restaurant quality
- Higher competition leading to improved service
- Stronger culinary culture

Data Visualization

To better understand patterns in the dataset, multiple visualizations were created.

Cuisine Distribution Graphs

Bar charts were used to show the frequency of different cuisines served across restaurants.

These visualizations helped quickly identify dominant cuisine categories.

City Distribution Charts

City-wise bar charts were generated to illustrate:

- Restaurant concentration
- Rating trends

These visuals made it easier to compare cities and identify regional patterns.

Key Insights from the Analysis

Several important insights were obtained from the dataset:

1. Dominance of Certain Cuisines

North Indian, Chinese, and Fast Food cuisines dominate the restaurant market in the dataset.

2. Urban Concentration

Restaurants are heavily concentrated in larger cities, which is expected due to higher population density and demand.

3. Rating Variability

Restaurant ratings vary significantly between cities, suggesting differences in service quality and customer expectations.

4. Multi-Cuisine Strategy

Many restaurants serve multiple cuisines, indicating a strategy to attract a wider range of customers.

Challenges Faced

During the analysis, a few challenges were encountered:

Multiple Cuisine Entries

Many restaurants listed multiple cuisines in a single field. This required splitting and restructuring the data before meaningful analysis could be performed.

Data Distribution Differences

Some cities had far more restaurants than others, which required careful interpretation when comparing ratings and averages.

Learning Outcomes

This project helped me strengthen my skills in:

- Exploratory Data Analysis
- Data cleaning and preprocessing
- Python data science libraries
- Data visualization
- Interpreting real-world datasets

It also improved my ability to translate raw data into meaningful insights that could be useful for business decision-making.

Conclusion

The restaurant dataset analysis provided valuable insights into cuisine popularity, city-wise restaurant distribution, and customer rating trends. The findings highlight the importance of understanding customer preferences and regional dining patterns when analyzing the restaurant industry.

This project demonstrates how data analysis techniques can be used to extract meaningful patterns from large datasets and present them in an interpretable way.

The skills applied in this project are fundamental for data science roles involving exploratory data analysis and business intelligence.

Future Improvements

Future extensions of this project could include:

- Machine learning models to predict restaurant ratings
- Sentiment analysis on restaurant reviews
- Price range vs rating analysis
- Clustering restaurants based on cuisine and ratings
- Building an interactive dashboard using Power BI or Tableau

These improvements would further enhance the analytical value of the dataset.

Author

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Project: Restaurant Data Analysis using Python

This project was completed as part of my learning journey in **data analysis and data science using Python**.